

Wooldridge Appendix A.1

The Summation Operator and Descriptive Statistics

A zero-background summary — written so that someone who has never seen this material can read it and understand it fully.

1. What is the Summation Operator (Σ)?

Imagine you are an accountant. You have five clients, and each one paid you a different amount this month:

Client 1 paid €100, Client 2 paid €200, Client 3 paid €150, Client 4 paid €300, Client 5 paid €250.

To find the total, you write: $100 + 200 + 150 + 300 + 250 = €1,000$. That works for 5 clients. But what about 500 clients? Or 10,000 bank branches in a dataset? Writing out every number becomes impossible. So mathematicians invented a shorthand — the **summation operator**, written as Σ (the Greek letter Sigma, which simply stands for "Sum").

The formula (A.1 in Wooldridge):

$$\Sigma(i=1 \text{ to } n) x_i = x_1 + x_2 + \dots + x_n$$

How to read it piece by piece:

x_i — the "i-th" value. A counter. $x_1 = 100$, $x_2 = 200$, $x_3 = 150$, etc. The little "i" just says "which client are we on?"

$i = 1$ (at the bottom of Σ) — start counting from client 1.

n (at the top of Σ) — keep going until the last one. Here $n = 5$.

Σ — add them all up. That is it. It is not new math — it is a shorter way to write addition when you have many items.

2. The Eight Properties of Summation

Property Sum.1 (A.2): Summing a constant

$$\Sigma(i=1 \text{ to } n) c = n \times c$$

If every client paid the same amount (a constant c), you do not need to add one by one. If all 5 clients paid €100, the total is just $5 \times 100 = €500$. Adding the same number n times is just multiplication.

Property Sum.2 (A.3): Pulling a constant out

$$\Sigma c x_i = c \times \Sigma x_i$$

Imagine the government taxes each client's payment at 10% ($c = 0.10$). You could calculate the tax on each client individually and then add them up. Or you could add up all payments first and then take 10% of the total — same answer either way. You can always "pull the constant out" of the summation.

Example: $0.10 \times 100 + 0.10 \times 200 + 0.10 \times 150 + 0.10 \times 300 + 0.10 \times 250 = 100$. Or: $0.10 \times 1,000 = 100$. Same answer.

Property Sum.3 (A.4): Splitting a combined sum

$$\Sigma (a x_i + b y_i) = a \Sigma x_i + b \Sigma y_i$$

Each client now has two numbers — revenue (x_i) and expenses (y_i). You can combine them client by client first and then add up, or sum each variable separately first and then combine. Same result. This is just Sum.2 extended to two variables.

⚠ THE WARNING — What Σ CANNOT Do

The sum of the ratios is NOT the ratio of the sums:

$$\Sigma(x_i/y_i) \neq (\Sigma x_i) / (\Sigma y_i)$$

Example: Client 1: revenue €100, expenses €50, ratio = 2.0. Client 2: revenue €300, expenses €200, ratio = 1.5. Sum of ratios = 3.5. But ratio of sums = 400/250 = 1.6. Completely different.

Same warning for squaring: the sum of squares $\neq$ the square of the sum. $(1^2 + 2^2) = 5$, but $(1+2)^2 = 9$.

3. The Average (Mean)

$$\bar{x} = (1/n) \times \Sigma x_i \quad (\text{formula A.5})$$

Add everything up, divide by the count. The symbol $\bar{x}$ ("x-bar") means "the average of x." You will see this bar notation on nearly every page of the textbook from now on.

The Median — a safer alternative

The average can be tricked by extreme values. If five houses cost €100k, €120k, €130k, €140k, and €2,000,000 — the average is €498k, which describes no actual house. The median — the middle value when sorted — is €130k, much more representative. That is why news reports use "median house price" or "median income." Rule: odd count $\rightarrow$ pick the middle; even count $\rightarrow$ average the two middle values.

4. Deviations Always Sum to Zero

$$\Sigma (x_i - \bar{x}) = 0 \quad (\text{formula A.6})$$

Take every value, subtract the average, add up all those differences — you always get exactly zero. Why? Think of the average as the balance point. Some values are above it (positive deviations), some are below (negative deviations). By definition, they cancel out perfectly.

Example: Five clients paid €100, €200, €150, €300, €250. Average = €200. Deviations: -100, 0, -50, +100, +50. Sum = 0. Always.

5. Sum of Squared Deviations

$$\Sigma (x_i - \bar{x})^2 = \Sigma x_i^2 - n(\bar{x})^2 \quad (\text{formula A.7})$$

Since deviations always sum to zero, raw deviations are useless for measuring how spread out data is — positive and negative cancel. Solution: square each deviation first (making everything positive), then add them up. The left side is the definition; the right side is a shortcut for computing the same thing faster. This number becomes variance when you divide by n . You will meet variance formally in Appendix B.

6. Cross-Product Deviations (Preview of Covariance)

$$\Sigma (x_i - \bar{x})(y_i - \bar{y}) = \Sigma x_i y_i - n(\bar{x})(\bar{y}) \quad (\text{formula A.8})$$

This extends A.7 to two variables. Instead of squaring, you multiply x 's deviation by y 's deviation. If x is a bank's loan growth and y is its profit growth, this formula measures whether they move together. When both are above average, the product is positive. When one is above and the other

below, the product is negative. The total tells you: do x and y tend to move in the same direction? This becomes covariance in Appendix B and is the backbone of regression in Chapter 2.

What Comes Next

Section A.2 (Linear Functions) introduces the single most important idea in the entire textbook: $y = \beta_0 + \beta_1 x$. Every regression you will ever run is built on this equation. That is the next session.

Wooldridge Appendix A.2 — Multivariate Linear Functions

Summary notes — 28 July 2026

1. The equation

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

With one variable (A.1), there was one intercept and one slope. With two variables, there is still one intercept (β_0), but now two slopes — one for each variable.

2. Slope = marginal effect

Each slope (β_1 or β_2) tells you how much y changes for a one-unit increase in that variable — the marginal effect. Analogy: $\text{total_cost} = 500 + 10 \times \text{units} + 2 \times \text{machine_hours}$. Here $\beta_1 = 10$ is the marginal effect of one more unit produced; $\beta_2 = 2$ is the marginal effect of one more machine hour.

3. Partial effect & ceteris paribus

β_1 is called the partial effect of x_1 on y , because it measures the effect of x_1 only while holding x_2 fixed (unchanged). This "holding everything else fixed" idea is called ceteris paribus. β_2 has the same interpretation in the other direction: the effect of x_2 , holding x_1 fixed.

4. Worked example — Demand for Compact Discs

$$\text{quantity} = 120 - 9.8 \times \text{price} + 0.03 \times \text{income}$$

$\beta_1 = -9.8$: holding income fixed, a \$1 price increase reduces quantity demanded by 9.8 units. $\beta_2 = 0.03$: holding price fixed, a \$1 income increase raises quantity demanded by 0.03 units. On a 2-D graph (income fixed at \$900), only price moves you along the line; a change in income shifts the whole line up or down instead.

5. A limitation to remember (from A.1)

In the earlier housing example ($\text{housing} = 164 + 0.27 \times \text{income}$), the intercept implies a family with \$0 income still spends \$164 on housing — unrealistic. Linear functions apply the same constant rate everywhere, including unrealistic extremes like $x = 0$. This is a known limitation, not a mistake — other function forms exist for cases where this matters.

6. Why this matters for the PhD

Every regression coefficient reported in an empirical paper is a partial effect — the effect of one variable, holding all other included controls fixed. When a referee questions whether a result is "really causal," they are asking whether the ceteris paribus condition truly holds, or whether an omitted variable is contaminating the estimate. This is the core logic behind "controlling for confounders" in applied econometrics.

Wooldridge Appendix A.3 — Proportions and Percentages

Standalone lesson — written to be read fresh, no prior context needed · 30 July 2026

Piece 1 — A proportion and a percentage are the same number, written two ways

A proportion is a decimal fraction — a number like .82, meaning "82 out of every 100." A percentage is that exact same value, just multiplied by 100 and written with a % sign. So .82 and 82% carry identical information; they are two different spellings of one number, the same way "1/2" and "0.5" are two spellings of the same value.

In one sentence: multiply a proportion by 100 to get its percentage — they always mean the same thing.

Piece 2 — Convert back to decimal form before you multiply

This is the single most common mistake people make with percentages: multiplying by the raw percentage number instead of its decimal form. If a sales tax is 6% and you spend \$200, the tax is NOT $200 \times 6 = \$1,200$ — that's obviously far too much. You must first convert 6% back into decimal form (divide by 100): $6\% = .06$. Then the correct tax is $200 \times .06 = \$12$.

Same logic with a savings example: a certificate of deposit paying 7.6% annual interest on a \$3,000 deposit does NOT earn $3,000 \times 7.6$. It earns $3,000 \times .076 = \$228$.

In one sentence: always convert a percentage back into decimal form (divide by 100) before multiplying — multiplying by the raw percentage number gives a wildly wrong answer.

Piece 3 — Watch for percentages reported sloppily in everyday writing

Sometimes a writer reports a decimal proportion, like .57, but calls it a "percentage" without converting it. Read literally, .57 as a percentage would mean 0.57% — barely over half of one percent. But the writer usually means 57%, a completely different, much larger number. The same error shows up in sports writing: "her hitting percentage was .372" really means 37.2%, not 0.372%.

In one sentence: when you see a decimal labeled as a "percentage" in casual writing, check whether the writer actually means the much larger true percentage value.

Piece 4 — Proportionate change: how much did a value move, relative to where it started?

Suppose someone's income rises from \$30,000 (call this the starting value, x_0) to \$36,000 (the new value, x_1). The proportionate change is: $(\text{new value} - \text{old value}) \div \text{old value} = (36,000 - 30,000) \div 30,000 = 6,000 \div 30,000 = .20$.

Why divide by the starting value at all? Because it makes the change comparable regardless of scale. A \$6,000 raise means something very different to a person earning \$30,000 than to a person earning \$300,000. Dividing by the starting value standardizes the change into a fair ratio, so changes at very different starting points can be compared honestly.

In one sentence: proportionate change = $(\text{change in } x) \div (\text{starting value of } x)$ — it turns a raw dollar change into a comparable ratio.

Piece 5 — Percentage change: the same idea, scaled to a percentage

Percentage change is just the proportionate change multiplied by 100 (Wooldridge equation A.15: $\% \Delta x = 100 \times (\Delta x / x_0)$). Example: income rises from \$30,000 to \$33,750. Proportionate change = $3,750 \div 30,000 = .125$. Multiply by 100 to get the percentage change: 12.5%.

A caution directly from the textbook: do not write ".125" and call it "the percentage change" — that repeats the Piece 3 mistake. The percentage change is 12.5 (or 12.5%), not .125.

Important clarification: .125 is a completely valid number to report — as long as it is labeled correctly. .125 is correct when called the proportionate change (that is exactly what proportionate change means, a raw ratio). It only becomes an error if that same number, .125, is labeled "the percentage change" instead — because "percentage change" specifically refers to the value after multiplying by 100. The number itself isn't wrong; the label attached to it has to match.

In one sentence: percentage change is the proportionate change multiplied by 100 — the same calculation as Piece 4, with one extra step, and the label you attach to the number must match which version (proportionate or percentage) you are reporting.

Piece 6 — Percentage POINT change vs. percentage change (the distinction that trips people up most)

This distinction only applies when the variable itself is already a percentage — for example, "the percentage of people with a college degree," not something like income or population.

Example: suppose 24% of a city's adults have a college education ($x_0 = 24$), and this rises to 30% ($x_1 = 30$).

- Percentage POINT change: simply $x_1 - x_0 = 30 - 24 = 6$ percentage points. Plain subtraction, nothing more.
- Percentage change: uses the Piece 5 formula — $(30 - 24) \div 24 \times 100 = 25\%$.

Notice: the exact same real-world change produces two very different numbers — 6 versus 25 — depending on which concept is being reported. The textbook's real-world example is the 1994 Michigan sales tax increase, which went from 4% to 6%. Supporters of the tax called this a "two percentage point increase" ($6 - 4 = 2$). Opponents called it "a 50% increase in the sales tax rate" ($((6 - 4) \div 4 \times 100 = 50\%)$). Both claims are technically correct — they are simply describing two different concepts, and each side chose the framing that made their position look most favorable.

One more detail: this percentage-point vs. percentage-change distinction only makes sense for a variable that is already measured as a percentage. For something like salary, it is meaningless to talk about a "percentage point change in salary," because salary itself is not a percentage — a change in salary can only be described in dollar terms or percentage-change terms, not percentage points.

In one sentence: when x is itself a percentage, percentage POINT change is simple subtraction, while percentage change divides that difference by the starting value — and reporting the wrong one can make the identical real-world change sound very different depending on who is describing it.

Quadratic Models

Use x^2 when the effect of x itself may change as x changes.

1. Core idea

A linear model forces the same marginal effect everywhere. A quadratic allows the slope to change with x , so the relationship can rise, flatten, peak, and reverse.

$$y = \beta_0 + \beta_1 x + \beta_2 x^2$$

Marginal effect

Formula

$$\partial y / \partial x = \beta_1 + 2\beta_2 x$$

Meaning

Do not interpret β_1 alone. The effect depends on the current value of x .

Turning point

Formula

$$x^* = -\beta_1 / (2\beta_2)$$

Meaning

This is where the slope equals zero: the peak for an inverted U, or the minimum for a U-shape.

2. Sign patterns

β_1	β_2	Typical shape
$\beta_1 > 0$	$\beta_2 < 0$	Inverted U: rises, peaks, then falls
$\beta_1 < 0$	$\beta_2 > 0$	U-shape: falls, bottoms out, then rises

Worked example: $y = 6 + 8x - 2x^2$

At $x = 1$

$$\text{Marginal effect} = 8 + 2(-2)(1) = 4.$$

Turning point

$$x^* = -8 / [2(-2)] = 2.$$

Empirical reading

The effect is positive below the peak, becomes zero at $x = 2$, and turns negative afterward.

Log Models

The transformation tells you whether the coefficient is in units or percentages.



Master rule

Always look first at which variable is logged: y, x, both, or neither. The logged variable is interpreted in percentage terms.

A. Log-log

$$\log(y) = \beta_0 + \beta_1 \log(x)$$

Interpretation: 1% increase in x → $\beta_1\%$ change in y

Memory: β_1 is the elasticity.

Example: $\beta_1 = 0.40$ means 1% more x is associated with about 0.40% more y.

B. Log-level

$$\log(y) = \beta_0 + \beta_1 x$$

Interpretation: 1 unit increase in x → $100\beta_1\%$ change in y

Memory: Unit → percent.

Example: $\beta_1 = -0.12$ means one more unit of x is associated with about a 12% fall in y.

C. Level-log

$$y = \beta_0 + \beta_1 \log(x)$$

Interpretation: 1% increase in x → $\beta_1/100$ units change in y

Memory: Percent → units.

Example: $\beta_1 = 30$ means 1% more x is associated with 0.30 more units of y.

Fast interpretation table

Model	Input → output	Coefficient rule
$\log(y)$ on $\log(x)$	% → %	β_1
$\log(y)$ on x	unit → %	$100\beta_1$
y on $\log(x)$	% → unit	$\beta_1/100$

Percentage Changes & Elasticity

Read log differences as growth rates and distinguish elasticity from slope.

1. Log differences approximate percentage changes



$$100 \times [\log(x_1) - \log(x_0)] \approx \% \Delta x$$

Small-change example: 40 → 41

Actual percentage change

$$(41 - 40) / 40 \times 100 = 2.5\%$$

Log approximation

$$[\log(41) - \log(40)] \times 100 \approx 2.47\%$$

Takeaway

For small changes, the approximation is very close.

Why the word “approximate” matters

Bigger change: 40 → 60

Actual change = 50%, but the log difference is about 40.55%.

Empirical habit

Treat $\Delta \log(x)$ as a percentage-growth measure mainly when changes are reasonably small.

2. Elasticity

$$\text{Elasticity} = \% \Delta y / \% \Delta x$$

Elasticity asks: if x changes by 1%, by what percentage does y change? In a log-log regression, the coefficient β_1 is directly the elasticity.

Slope is not elasticity

Slope

A unit change in x produces a unit change in y .

Elasticity

A percentage change in x produces a percentage change in y .

Key point

Even on a straight line with constant slope, elasticity can change with the starting values of x and y .

Empirical reading example: $\log(\text{Lending}) = 3 - 0.6\log(\text{Deposits})$

Elasticity

-0.6

Interpretation

A 1% increase in deposits is associated with approximately a 0.6% decrease in lending.

Exponential Function + Final Workflow

Use $\exp()$ to convert a predicted log outcome back to its original units.

1. $\exp()$ is the inverse of $\log()$

$$\log(\exp(x)) = x \quad \text{and} \quad \exp(\log(x)) = x$$

If your regression predicts $\log(y)$, exponentiating returns the prediction to the original units of y .

Prediction example

Step 1

Suppose predicted $\log(y) = 3$.

Step 2

$y = \exp(3) \approx 20.09$.

Meaning

The model prediction is 3 in log units and about 20.09 in the original units of y .

Do not confuse prediction with coefficient interpretation

Coefficient interpretation

For $\log(y) = \beta_0 + \beta_1 x$, one extra unit of x changes y by about $100\beta_1\%$.

Prediction

Use $\exp()$ when you want the predicted y itself back in original units.

2. Four-batch empirical checklist

Quadratic

- 1 Need a changing marginal effect? Include x and x^2 ; report marginal effects and the turning point when relevant.

Logs

- 2 Identify which variable is logged before interpreting β .

Growth / elasticity

- 3 $100\Delta\log(x)$ approximates $\% \Delta x$; in a log-log model β is elasticity.

Prediction

- 4 If the dependent variable is logged and you need y in original units, apply $\exp()$ to the predicted log outcome.

FINAL STUDY RULE

Do not memorize a coefficient interpretation before identifying the functional form. The functional form tells you how the coefficient must be read.

WOOLDRIDGE A.5

Differential Calculus for Empirical Research

The complete step-by-step lesson taught in our chat

Derivatives | Partial derivatives | Quadratic terms | Interactions | First-order conditions

Purpose of this file

This document reproduces the lesson as it was taught: one idea at a time, with the same examples, interpretations, mistakes, and corrections. It is not a restyled version of the original summary.

Prepared 3 August 2026

How to use this lesson

Read the batches in order. Each batch adds only one new idea. The main goal is not to memorize calculus rules. The goal is to understand what regression coefficients, quadratic terms, interaction terms, and turning points mean in empirical research.

- A derivative is a marginal effect: how the predicted outcome changes when one explanatory variable changes.
- A partial derivative means that other explanatory variables are held constant while one variable changes.
- A quadratic term makes the marginal effect depend on the current level of the variable.
- An interaction term makes the effect of one variable depend on another variable.
- A first-order condition locates a turning point by setting the derivative equal to zero.

Learning rule used in the chat

When an answer was wrong, the mistake was corrected once, the fix was explained, and the lesson continued. The correction is included where it mattered.

BATCH 1 | A derivative is the coefficient in a simple linear model

Start with a simple wage model:

$$\text{wage} = 3 + 0.5\text{education}$$

The derivative of wage with respect to education is:

$$d\text{wage} / d\text{education} = 0.5$$

Interpretation: one additional year of education is associated with a 0.5-unit increase in wage.

In a simple linear model, the coefficient on x is also the marginal effect of x :

$$y = \beta_0 + \beta_1 x \rightarrow dy / dx = \beta_1$$

Empirical research example

$$\text{loan_spread} = 200 - 15\text{AI}$$

If AI is a binary indicator, the coefficient -15 means that loans from AI-adopting banks have predicted spreads that are 15 basis points lower, on average. Because AI changes from 0 to 1, this is more precisely a discrete difference, but the interpretation still concerns the change in the predicted outcome.

The first correction from the lesson

$$\text{small_business_lending} = 50 + 4\text{deposit_growth}$$

- 50 is the constant or starting point.
- 4 is the coefficient and the derivative.

Therefore, when deposit growth increases by one unit, predicted small-business lending increases by 4 units. If deposit growth increases by two units, lending increases by 8 units.

Core rule

In a simple linear model, the regression coefficient is the marginal effect. Always state the units of both variables.

BATCH 2 | Partial derivatives: freeze one variable and examine the other

$$\text{lending} = 20 + 4\text{deposit_growth} + 2\text{bank_size}$$

There are now two explanatory variables. To study deposit growth, keep bank size unchanged:

$$\partial \text{lending} / \partial \text{deposit_growth} = 4$$

Interpretation: **holding bank size constant**, a one-unit increase in deposit growth increases predicted lending by 4 units.

To study bank size, keep deposit growth unchanged:

$$\partial \text{lending} / \partial \text{bank_size} = 2$$

Interpretation: holding deposit growth constant, a one-unit increase in bank size increases predicted lending by 2 units.

Important distinction

Holding a variable constant does not mean setting it equal to zero. It means its value does not move while the other variable changes.

Connection to empirical research

$$\text{loan_spread} = 200 - 15\text{AI} + 6\text{risk} + 3\text{loan_size}$$

The coefficient -15 means: holding borrower risk and loan size constant, AI adoption is associated with a 15-basis-point lower loan spread.

$$\text{wage} = 5 + 2\text{education} + 3\text{experience}$$

The coefficient 3 means: holding education constant, one additional year of experience increases predicted wage by 3 units. Two additional years increase it by 6 units.

Wording correction

Say “wage increases by 3 units,” not “by 3 coefficients.” The coefficient is the number used to describe the change.

BATCH 3 | Quadratic terms: the effect changes with the level of x

$$\text{wage} = 5 + 3\text{experience} - 0.05\text{experience}^2$$

The squared term means that the effect of experience is not constant. Differentiate term by term:

- 5 contributes 0 because it is constant.
- 3experience contributes 3.
- -0.05experience^2 contributes -0.10experience .

$$\partial \text{wage} / \partial \text{experience} = 3 - 0.10\text{experience}$$

Evaluate the marginal effect at different experience levels

$$\text{At experience} = 5: 3 - 0.10(5) = 2.5$$

One additional year of experience increases predicted wage by 2.5 units.

$$\text{At experience} = 20: 3 - 0.10(20) = 1$$

One additional year increases predicted wage by only 1 unit.

$$\text{At experience} = 30: 3 - 0.10(30) = 0$$

At 30 years, the fitted model reaches its turning point: another year has a zero marginal effect.

Diminishing marginal returns

Experience still raises wages before the turning point, but each additional year raises wages by less than the previous year.

Checkpoint from the lesson

$$\text{productivity} = 10 + 4\text{training} - 0.20\text{training}^2$$

$$\partial \text{productivity} / \partial \text{training} = 4 - 0.40\text{training}$$

$$\text{At training} = 5: 4 - 0.40(5) = 2$$

The answer was 2: when training equals 5, one additional unit of training increases predicted productivity by 2 units.

BATCH 4 | Interaction terms: the effect of one variable depends on another

$$\text{wage} = 5 + 2\text{education} + 3\text{experience} + 0.5(\text{education} \times \text{experience})$$

The interaction term says that education and experience do not operate independently. The effect of experience changes with education.

Freeze education and differentiate with respect to experience:

$$\partial \text{wage} / \partial \text{experience} = 3 + 0.5\text{education}$$

$$\text{At education} = 4: 3 + 0.5(4) = 5$$

$$\text{At education} = 10: 3 + 0.5(10) = 8$$

More education makes each additional year of experience more valuable. The interaction coefficient 0.5 means that one extra year of education raises the marginal effect of experience by 0.5 units.

Empirical research example

$$\text{loan_spread} = 200 - 10\text{bankAI} - 5\text{firmAI} - 8(\text{bankAI} \times \text{firmAI})$$

- For a non-AI firm, the effect of bank AI is -10 basis points.
- For an AI firm, the effect is $-10 - 8 = -18$ basis points.

The interaction tells us that the relationship between bank AI adoption and loan spreads differs depending on whether the borrower also uses AI.

Correction from the lesson

$$\text{productivity} = 10 + 2\text{training} + 3\text{technology} + 0.4(\text{training} \times \text{technology})$$

$$\partial \text{productivity} / \partial \text{training} = 2 + 0.4\text{technology}$$

$$\text{At technology} = 5: 2 + 0.4(5) = 2 + 2 = 4$$

The mistake and the fix

The answer 2 included only the interaction contribution. The complete marginal effect also includes the base training effect of 2, so the correct answer is 4.

BATCH 5 | First-order conditions with one variable

$$\text{profit} = 100 + 20x - x^2$$

The function first rises, reaches a turning point, and then falls. Take the derivative:



$$d\text{profit} / dx = 20 - 2x$$

At the turning point, the slope is zero. Set the derivative equal to zero:

$$20 - 2x = 0 \rightarrow x = 10$$

- Before $x = 10$, the derivative is positive, so increasing x raises profit.
- At $x = 10$, the derivative equals zero.
- After $x = 10$, the derivative is negative, so increasing x lowers profit.

Therefore, $x = 10$ is the profit-maximizing level for this downward-opening quadratic.

Empirical-research connection

$$\text{wage} = 5 + 3\text{experience} - 0.05\text{experience}^2$$

$$d\text{wage} / d\text{experience} = 3 - 0.10\text{experience}$$

$$3 - 0.10\text{experience} = 0 \rightarrow \text{experience} = 30$$

The fitted quadratic model predicts that wages reach their turning point at 30 years of experience.

First-order condition

Take the derivative, set it equal to zero, and solve for the turning point.

BATCH 6 | First-order conditions with two variables

$$\text{profit} = 20L + 16K - L^2 - K^2$$

Profit depends on labor L and capital K , so both partial derivatives must equal zero at the same time.

Differentiate with respect to labor

$$\partial \text{profit} / \partial L = 20 - 2L$$

$$20 - 2L = 0 \rightarrow L = 10$$

Differentiate with respect to capital

$$\partial \text{profit} / \partial K = 16 - 2K$$

$$16 - 2K = 0 \rightarrow K = 8$$

The turning point occurs at $L = 10$ and $K = 8$. At that point, a small increase in either labor or capital has a zero marginal effect on profit.

Why does $-K^2$ disappear when differentiating with respect to L ?

When differentiating with respect to L , K is frozen. This does not mean K^2 equals zero. It means K^2 does not change when L changes, so it behaves like a constant.

If $K = 5$, then $-K^2 = -25$

As L changes, -25 remains -25 . The derivative of a constant with respect to L is zero. Therefore, the $-K^2$ term contributes zero to $\partial \text{profit} / \partial L$.

Exact correction

We are not saying $K^2 = 0$. We are saying $\partial(-K^2) / \partial L = 0$ because K is held constant while L changes.

Limitation of the first-order condition

A zero derivative identifies a stationary or turning point, but zero alone does not always prove that the point is a maximum. It could also be a minimum or another stationary point. Second derivatives provide the formal classification, although this appendix does not require that step.

BATCH 7 | A quadratic term and an interaction term together

Now combine the two ideas in Wooldridge's wage equation:

$$\text{wage} = 3.10 + 0.41\text{educ} + 0.19\text{exper} - 0.004\text{exper}^2 + 0.007(\text{educ} \times \text{exper})$$

To find the effect of experience, freeze education and differentiate term by term:

- 3.10 contributes 0 .
- 0.41educ contributes 0 because education is frozen.
- 0.19exper contributes 0.19 .
- -0.004exper^2 contributes -0.008exper .
- $0.007\text{educ} \times \text{exper}$ contributes 0.007educ .

$$\partial \text{wage} / \partial \text{exper} = 0.19 - 0.008\text{exper} + 0.007\text{educ}$$

Meaning of each part

- -0.008exper : the return to experience diminishes as existing experience rises.
- $+0.007\text{educ}$: more education makes an additional year of experience more valuable.

Numerical example

$$\text{exper} = 10, \text{educ} = 12$$

$$0.19 - 0.008(10) + 0.007(12) = 0.194$$

For a worker with 10 years of experience and 12 years of education, one additional year of experience increases predicted wage by approximately 0.194 units, holding education constant.

Combined interpretation

The payoff from experience becomes smaller as experience accumulates, but it becomes larger when education is higher.

BATCH 8 | Turning points when an interaction exists

Start from the complete marginal effect of experience:

$$\partial \text{wage} / \partial \text{exper} = 0.19 - 0.008\text{exper} + 0.007\text{educ}$$

Set it equal to zero and solve for experience:

$$0.19 - 0.008\text{exper} + 0.007\text{educ} = 0$$

$$\text{exper}^* = (0.19 + 0.007\text{educ}) / 0.008$$

The star indicates the experience value at the turning point.

Different education levels produce different turning points

$$\text{At educ} = 12: \text{exper}^* = (0.19 + 0.007(12)) / 0.008 = 34.25$$

$$\text{At educ} = 16: \text{exper}^* = (0.19 + 0.007(16)) / 0.008 = 37.75$$

The fitted model predicts a later wage peak for a more highly educated worker because education raises the marginal return to experience.

What “no universal turning point” means

You cannot report one experience level and imply that it applies to everyone. The turning-point formula contains education, so the answer changes when education changes.

Incorrect reporting

“Wages peak after 35 years of experience.” This wrongly suggests that the peak is identical for every education level.

Correct reporting

“The predicted experience level at which wages peak depends on education: approximately 34 years at 12 years of education and 38 years at 16 years of education.”

Without the interaction term, education would not appear in the turning-point formula, and the model could have one common turning point.

BATCH 9 | How to report the result in empirical research

$$\text{wage} = 3.10 + 0.41\text{educ} + 0.19\text{exper} - 0.004\text{exper}^2 + 0.007(\text{educ} \times \text{exper})$$

Correct interpretation

Research-paper wording

Holding education constant, experience has a positive but diminishing association with wages. However, the positive interaction between education and experience indicates that the marginal return to experience is larger for more highly educated workers.

What not to write

Incomplete statement

“One additional year of experience increases wages by 0.19.”

This statement is incomplete because 0.19 is only the coefficient on the linear experience term. The complete marginal effect also depends on existing experience and education:

$$\partial \text{wage} / \partial \text{exper} = 0.19 - 0.008\text{exper} + 0.007\text{educ}$$

Central empirical rule

When a variable appears in a quadratic term or an interaction term, do not interpret its individual linear coefficient by itself. Calculate and interpret the complete marginal effect.

The complete logic in one page

Model feature	What changes	What to interpret
Linear term only	The marginal effect is constant.	Interpret the coefficient directly.
Several variables	Other variables are frozen.	Use a partial derivative and say "holding other factors constant."
Quadratic term	The effect changes with the current level of the same variable.	Evaluate the marginal effect at meaningful values.
Interaction term	The effect of one variable changes with another variable.	Combine the base coefficient and interaction contribution.
First-order condition	The marginal effect becomes zero at a turning point.	Set the derivative equal to zero and solve.
Quadratic plus interaction	Both the marginal effect and the turning point depend on another variable.	Report conditional effects and conditional turning points.

Final takeaway

Differential calculus in empirical research is mainly a language for reading models correctly. It tells you whether an effect is constant, changes with the variable itself, changes with another variable, or reaches a turning point.

Wooldridge Appendix B.1 — Random Variables and Their Probability Distributions

Standalone lesson — intuition only, written to be read fresh · 9 August 2026

Piece 1 — What is a random variable, really?

Think about flipping a coin 10 times and counting how many times it lands on heads. Before you actually do it, you don't know the answer — it could be 3, it could be 7, anything from 0 to 10. Once you actually flip it, you get one specific number.

In one sentence: a random variable is a quantity that's uncertain beforehand, but takes a specific value once the real-world event actually happens.

Piece 2 — Why this matters: the airline seat example

An airline has 100 seats. If they only accept 100 reservations, some people won't show up and seats go empty. So airlines often overbook — accept more than 100 reservations, betting some people won't show. Risk: if more than 100 actually show up, the airline has to pay people to give up seats.

In one sentence: probability exists to help answer real business questions like "how many reservations to accept," where the actual outcome is a random variable — unknown in advance, but critical to plan around. The same shape of problem shows up in research: how many borrowers will default, how many loans will get refinanced.

Piece 3 — Discrete vs. continuous

Discrete means the possible outcomes are a countable list — like a coin flip result (0 through 10 heads). Continuous means the outcomes could be any number within a range — like an exact stock price or exact income — because there are infinitely many possible numbers between any two points.

In one sentence: discrete means you can count the possibilities, even if there are many; continuous means the possibilities are infinite and unlistable, like any point along a smooth line.

Piece 4 — The Bernoulli variable

A Bernoulli variable can only be one of two values: 1 or 0 — "success" or "failure." Like a single light switch: on or off, nothing in between.

Example: for a customer with a flight reservation, $X = 1$ if they show up, $X = 0$ if they don't. The probability of showing up is called θ (theta) — a symbol for "whatever that probability actually is."

Named after Jacob Bernoulli, a Swiss mathematician (1600s–1700s).

Connection to econometrics: a Bernoulli random variable is the probability-theory concept for something genuinely uncertain. A dummy variable (or indicator variable) is the econometrics term for a 0/1 variable used inside a regression (e.g. "is this borrower a fintech customer?" 1 = yes, 0 = no). Same underlying 0/1 object — Bernoulli when discussing pure probability, "dummy variable" once it's an input inside a regression equation.

In one sentence: a Bernoulli variable is the simplest possible random variable — a yes/no switch with probability θ attached — and it's the same mathematical object as the "dummy variable" used constantly in regressions.

Piece 5 — The PDF (probability density function)

The PDF is a list pairing each possible outcome with its probability.

Example — a basketball player attempts 2 free throws. X = number made (0, 1, or 2):

$$f(0) = .20 \quad f(1) = .44 \quad f(2) = .36$$

These always sum to 1.00 ($.20 + .44 + .36 = 1.00$), because all probabilities across every possible outcome must add up to exactly 100% — something has to happen.

In one sentence: the PDF is a probability lookup table for each exact outcome, and those probabilities always sum to 1.

Piece 6 — The CDF (cumulative distribution function)

Instead of "what's the probability of exactly this value," the CDF answers "what's the probability of this value or less?"

Using the same free throw example:

$$F(1) = P(X \leq 1) = f(0) + f(1) = .20 + .44 = .64$$

A 64% chance of making 1 or fewer free throws.

In one sentence: the CDF answers "what's the probability of getting this value or lower," building directly on the PDF by adding up probabilities as you go.

Piece 7 — Why this section matters going forward

Random variables, discrete vs. continuous, PDFs, and CDFs are the basic language for describing uncertainty in data. Later, when studying standard errors, confidence intervals, and hypothesis tests, all of those build directly on these ideas: a regression coefficient itself is treated as a random variable, and the CDF (specifically the normal distribution's) is used to calculate confidence in results.

Wooldridge Appendix B.2 — Joint Distributions, Conditional Distributions, and Independence

Standalone lesson — written to be read fresh, no prior context needed · 11 August 2026

Piece 1 — Why we need this: moving beyond one variable at a time

B.1 was about a single random variable. In real economics, we usually care about two things happening together. Example: an airline isn't just interested in "will this person show up" — they might want "will this person show up AND are they a business traveler?" That's a joint probability — the chance of two specific things both being true at once.

In one sentence: a joint probability answers "what's the chance these two specific things both happen," not just one thing on its own.

Piece 2 — The free throw example: independence in action

A basketball player shoots two free throws. X = made the 1st shot (1 or 0), Y = made the 2nd shot (1 or 0). She's an 80% shooter, so $P(X=1) = P(Y=1) = .8$.

If the two shots are independent (one shot doesn't affect the other):

$$P(X=1, Y=1) = P(X=1) \times P(Y=1) = .8 \times .8 = .64$$

In one sentence: when two events are independent, the chance of both happening is simply their individual probabilities multiplied together.

Piece 3 — What "independent" actually means

Two variables are independent if knowing the outcome of one tells you nothing about the other. If shots are independent, missing the first shot doesn't change your 80% chance on the second — you're still exactly 80%, no matter what happened before. This is exactly why the multiplication trick in Piece 2 works.

In one sentence: independence means "the outcome of one thing gives you zero information about the other" — that's exactly what allows the simple multiplication shortcut.

Piece 4 — Extending beyond just two variables

The same idea works for many variables at once. Real example: an airline accepts 120 reservations. For each person, there's a Bernoulli variable (show up = 1, no-show = 0). Assuming all 120 people's decisions are independent lets you calculate the total number who'll show up.

Honest limitation: independence is only an approximation. In reality, people sometimes travel in groups — if one cancels, others might too. Treating them as independent is a useful simplification, not a perfect truth.

In one sentence: independence extends to many variables at once, and real researchers often use it as a useful approximation, even when it's not perfectly true.

Piece 5 — The binomial distribution: counting up "successes"

Remember the airline overbooking example from B.2? The airline accepts 120 reservations. For each person, there's a yes/no: **will they show up?** That's 120 separate Bernoulli trials — one per person.

The binomial distribution answers: **out of those 120 people, what's the probability that more than 100 actually show up?**

In one sentence: binomial takes many individual yes/no outcomes (120 people, each either showing up or not) and tells you the probability of a specific total count — like "more than 100 show up."

A **binomial distribution** is a statistical probability model that helps you figure out the likelihood of getting a specific number of "successes" out of a fixed number of yes/no or pass/fail tries.

The 4 Key Conditions

For a situation to use a binomial distribution, it must meet these four rules:

- **Fixed number of trials:** You repeat an action a set number of times (e.g., rolling a die 20 times).
- **Two possible outcomes:** Every trial results in only two options, usually labeled **success** or **failure** (e.g., passing or failing a test).
- **Constant probability:** Means the chance of success is the exact same for every trial (e.g., a 50% chance to guess a true/false question right, whether it's question 1 or question 10).
- **Independence:** Means guessing question 1 doesn't change or give you a clue about the answer to question 2.

Piece 6 — Conditional distributions: when independence breaks down

Independence means one outcome tells you nothing about the other — but often that's not true. A conditional distribution captures the probability of Y, GIVEN that X already happened.

Free throw example, now with dependence (Example B.2): if the player makes the first shot, their chance of making the second is 85%. If they miss the first shot, their chance on the second drops to 70%.

In one sentence: a conditional distribution answers "given that I already know X happened, what's the probability of Y now" — and this only matters when X and Y aren't independent.

Piece 7 — Using the conditional probability to find the joint probability

Given $P(\text{making 2nd shot} \mid \text{made 1st shot}) = .85$, and $P(\text{making 1st shot}) = .8$, the joint probability of making BOTH shots (even though they're not independent):

$$P(X=1, Y=1) = P(Y=1 \mid X=1) \times P(X=1) = .85 \times .8 = .68$$

In one sentence: even when two events aren't independent, their joint probability can still be found — by multiplying the conditional probability by the base probability, instead of multiplying two separate independent probabilities.

Extra worked example — independent vs. dependent, side by side

Scenario: X = passed a first exam (1 or 0), Y = passed a second exam (1 or 0). $P(X=1) = .7$ and $P(Y=1) = .7$ on their own.

Case 1 — INDEPENDENT (two unrelated subjects, e.g. French and Physics). Passing French tells you nothing about Physics:

$$P(X=1, Y=1) = P(X=1) \times P(Y=1) = .7 \times .7 = .49$$

Case 2 — DEPENDENT (the second exam builds directly on the first, e.g. Calculus 1 then Calculus 2). Passing exam 1 genuinely boosts the odds on exam 2. Say $P(Y=1 \mid X=1) = .90$:

$$P(X=1, Y=1) = P(Y=1 \mid X=1) \times P(X=1) = .90 \times .7 = .63$$

Note: .63 is bigger than .49 — because the dependence between the two exams is positive (doing well on the first genuinely helps the second), the joint probability of passing both went up compared to treating them as unrelated.

In one sentence: independent events (French/Physics) get multiplied straight across; dependent events (Calc 1/Calc 2) require the conditional probability instead, because knowing one outcome genuinely shifts the odds on the other, changing the final joint answer.

Wooldridge Appendix B.3 — Complete

Features of Probability Distributions · 12 August 2026 · includes original 10 pieces + Average vs. Expected Value supplement

Piece 1 — Expected Value: Definition and the "Curious" Example

A pdf tells you how likely each outcome is. Expected value takes that and produces one single number summarizing the "center" — a weighted average of every possible outcome, weighted by how likely each one is.

Example: X can be -1 , 0 , or 2 , with probabilities $1/8$, $1/2$, and $3/8$.

$$E(X) = (-1)(1/8) + (0)(1/2) + (2)(3/8) = 5/8$$

The curious part: $5/8$ is NOT a possible value of X — X can only ever actually be -1 , 0 , or 2 , never $5/8$. Expected value isn't "the most likely result" — it's a mathematical balance point, like a seesaw balancing at a spot where nobody's actually sitting.

In one sentence: expected value is a weighted average of all possible outcomes, and it can land on a number that's never actually one of the real possible outcomes.

Piece 2 — Functions of Random Variables: $E[g(X)]$

Continuing the same example, let $g(X) = X^2$:

$$E(X^2) = (-1)^2(1/8) + (0)^2(1/2) + (2)^2(3/8) = 13/8$$

Compare this to squaring the Piece 1 answer instead:

$$[E(X)]^2 = (5/8)^2 = 25/64$$

These are different numbers. You cannot swap the order of "apply the function" and "take the expected value" — squaring first-then-averaging gives a different answer than averaging first-then-squaring.

In one sentence: $E(X^2)$ and $[E(X)]^2$ are generally different numbers — a real trap if you assume you can swap the order.

Piece 3 — Properties of Expected Value (E.1, E.2, E.3)

$$\text{Property E.1: } E(c) = c$$

The expected value of a plain constant is just that constant.

$$\text{Property E.2: } E(aX + b) = aE(X) + b$$

If you scale a random variable and shift it, its expected value scales and shifts the exact same way.

Example — Celsius to Fahrenheit. X = temperature in Celsius, $E(X) = 25$. $Y = 32 + (9/5)X$.

$$E(Y) = 32 + (9/5)(25) = 32 + 45 = 77$$

$$\text{Property E.3: } E(a_1X_1 + a_2X_2 + \dots) = a_1E(X_1) + a_2E(X_2) + \dots$$

The expected value of a sum is the sum of the expected values.

In one sentence: these three properties let you calculate expected values quickly using simple algebra, without redoing the full weighted-average calculation every time.

Piece 4 — Applying the Properties: Pizza, Binomial, Airline

Example 1 — Pizza revenue (Property E.3). Expected daily sales: $E(X_1)=25$, $E(X_2)=57$, $E(X_3)=40$, priced at \$5.50, \$7.60, \$9.15.

$$\text{Expected revenue} = 5.50(25) + 7.60(57) + 9.15(40) = \$936.70$$

Example 2 — The binomial's expected value. A binomial variable is many independent Bernoulli trials added together. Using Property E.3:

$$E(X) = n\theta \quad (n = \text{number of trials}, \theta = \text{success probability})$$

Applying to the airline: 120 reservations, $\theta = .85$ chance of showing up.

$$E(X) = 120 \times .85 = 102$$

A real problem for the airline: the plane only has 100 seats, but the expected number showing up is already 102. The text notes the airline should build a full profit function — accounting for revenue per seat sold and the cost of bumping passengers — to find the reservation number that maximizes expected profit.

In one sentence: Property E.3 gives quick expected-value calculations, and applying it to the airline shows 120 reservations already expects more people (102) than seats (100).

Piece 5 — The Median: A Second Way to Measure "Center"

The median splits a distribution exactly in half — half the probability lies below it, half above.

Odd number of values: line them up, pick the middle one.

$$X = \{-4, 0, 2, 8, 10, 13, 17\} \rightarrow \text{median} = 8 \text{ (the 4th value)}$$

Even number of values: two middle values, sometimes averaged.

$$X = \{-5, 3, 9, 17\} \rightarrow \text{middle values } 3 \text{ and } 9 \rightarrow \text{median} = 6$$

Median and expected value are usually different numbers, and neither is "more correct" — both are valid ways to describe the center. They only match when the distribution is symmetric.

In one sentence: the median is a second, equally valid way to describe a distribution's center, and it usually differs from the expected value except when the distribution is symmetric.

Piece 6 — Variance: Measuring Spread

The idea: for each outcome, measure the distance from the mean ($X - \mu$), square it, then take the expected value of that squared distance:

$$\text{Var}(X) = E[(X - \mu)^2]$$

Worked example, continuing $X = -1, 0, 2$ with probabilities $1/8, 1/2, 3/8$, and $\mu = E(X) = 5/8$:

$$\text{AT } X=-1 \text{ distance} = -1 - 5/8 = -13/8, \text{ squared} = 169/64$$

$$\text{AT } X=0 \text{ distance} = 0 - 5/8 = -5/8, \text{ squared} = 25/64$$

$$\text{AT } X=2 \text{ distance} = 2 - 5/8 = 11/8, \text{ squared} = 121/64$$

Weighted average of these squared distances:

$$\text{Var}(X) = (169/64)(1/8) + (25/64)(1/2) + (121/64)(3/8) = 632/512 \approx 1.234$$

Check using the shortcut formula B.24:

$$\text{Var}(X) = E(X^2) - \mu^2 = 13/8 - (5/8)^2 = 1.625 - 0.391 = 1.234$$

In one sentence: variance takes the average squared distance between each outcome and the mean, giving one number that summarizes how spread out a distribution is.

Piece 7 — Properties of Variance (VAR.1, VAR.2)

Property VAR.1: $\text{Var}(X) = 0$ only if X is a constant

Property VAR.2: $\text{Var}(aX + b) = a^2 \text{Var}(X)$

Two things different from expected value's Property E.2: (1) adding a constant (b) does NOT change the variance at all; (2) multiplying by a constant (a) squares its effect on variance — not $\times a$, but $\times a^2$.

Example — Celsius to Fahrenheit again, $Y = 32 + (9/5)X$:

$$\text{Var}(Y) = (9/5)^2 \times \text{Var}(X) = (81/25) \times \text{Var}(X)$$

In one sentence: shifting a variable never changes its variance, but scaling it multiplies the variance by the square of that scale factor.

Piece 8 — Standard Deviation: More Natural Than Variance

Variance is in squared units — not intuitive. Standard deviation fixes this by taking the square root:

$$\text{sd}(X) = \sqrt{\text{Var}(X)}$$

Example: X = income in thousands, $E(X)=20$, $\text{sd}(X)=6$. Convert to plain dollars, $Y = 1000X$:

$$E(Y) = 1000 \times 20 = 20,000 \quad \text{sd}(Y) = 1000 \times 6 = 6,000$$

Both scale by exactly 1000 — simple and intuitive. Using variance instead, $\text{Var}(Y)$ would jump by a factor of one million (1000^2), not 1000.

In one sentence: standard deviation is the square root of variance, restoring normal units and scaling in a simple, intuitive way that variance doesn't.

Piece 9 — Standardizing a Random Variable: The Z-Score

The problem: two different variables (an 85/100 exam score and a 180cm height) can't be compared using raw numbers alone.

The fix: measure how far each value is from its own group's average, relative to how spread out that group normally is. Average exam score 70, typical spread 10 $\rightarrow$ 85 is 1.5 typical-spreads above average. Average height 175cm, typical spread 10cm $\rightarrow$ 180cm is only 0.5 typical-spreads above average. Even though 180 is a bigger raw number, the exam score is actually more unusual.

The formula is just this idea in symbols:

$$Z = (X - \mu) / \sigma$$

Standardizing a variable always produces:

$$E(Z) = 0 \quad \text{Var}(Z) = 1 \quad \text{sd}(Z) = 1$$

Example: $E(X)=2$, $\text{Var}(X)=9$ (so $\sigma=3$). Then $Z = (X-2)/3$ automatically has mean 0 and variance 1, regardless of X 's original values.

In one sentence: standardizing always produces mean 0 and standard deviation 1, letting completely different variables be compared on the same common scale.

Piece 10 — Skewness and Kurtosis (Brief)

Skewness — is the distribution lopsided? Cube the standardized variable Z instead of squaring it:

$$\text{Skewness} = E(Z^3)$$

Cubing keeps the sign. If perfectly symmetric, positive and negative cubed distances cancel exactly, giving $E(Z^3)=0$. If skewed (e.g. most incomes near \$40,000 with a few billionaires pulling the average up), they don't cancel.

Kurtosis — how thick are the extreme tails? Raise Z to the 4th power instead:

$$\text{Kurtosis} = E(Z^4)$$

Since Z^4 can never be negative, this is always positive. Higher kurtosis means more frequent extreme outliers, "thicker tails."

In one sentence: skewness ($E(Z^3)$) detects lopsidedness, while kurtosis ($E(Z^4)$) measures how often extreme, far-from-average values show up.

Supplement — Average vs. Expected Value: What's the Real Difference?

The core distinction: average is calculated from real data you already collected. Expected value is calculated from probabilities, before you have any real data yet.

Example 1 (Average): 5 students already took a test, scoring 70, 80, 90, 60, 100.

$$\text{Average} = (70+80+90+60+100) / 5 = 80$$

Real numbers already in hand — just add and divide.

Example 2 (Expected value): A game show has 3 possible prizes, nobody has played yet: \$0 (60% chance), \$100 (30% chance), \$1000 (10% chance).

$$E(\text{prize}) = (0)(.6) + (100)(.3) + (1000)(.1) = \$130$$

Only probabilities are known — no real outcomes exist yet.

The quick test to tell them apart: ask "do I already have real, collected numbers, or do I only have probabilities of what might happen?" Real numbers in hand → average. Only probabilities, nothing has happened yet → expected value.

What does \$130 actually mean, if it's not one of the possible prizes? It does NOT mean any single play gives you \$130 — look at the actual prizes: \$0, \$100, or \$1000. \$130 isn't even a possible outcome of one game.

Imagine playing this game 1,000 times. Based on the probabilities, roughly 600 times you'd win \$0, 300 times you'd win \$100, 100 times you'd win \$1000.

$$\text{Total winnings} = (600)(0) + (300)(100) + (100)(1000) = \$130,000$$

$$\text{Average per game} = \$130,000 / 1,000 \text{ games} = \$130$$

In one sentence: \$130 doesn't mean any single play gives you \$130 — it means if you played this exact game 1,000 times, your total winnings divided by 1,000 games would come out to about \$130 per game, on average.

~~**How average and expected value connect (the archery picture):** expected value is the bullseye — the true, exact center you're aiming at, but you can never see it directly. Average is where your arrows actually landed — your real, calculated data, your best estimate of where the bullseye is. If you collected a huge amount of real data and calculated the plain average, it would get closer and closer to matching the expected value — they converge to the same number, since they're two tools for two different situations: one for when you have raw data in hand, one for when you only know the probabilities.~~

Numeric proof: flip the earlier example (values -1, 0, 2 with probabilities 1/8, 1/2, 3/8) 800 times. Roughly 100 times you'd get -1, 400 times you'd get 0, 300 times you'd get 2.

$$\text{Sample average} = [(-1)(100) + (0)(400) + (2)(300)] / 800 = 500/800 = 5/8$$

Exactly matching the expected value from Piece 1.

In empirical research, which do you actually use? Almost always the sample average — because empirical research always starts from real, collected data, not just known probabilities. "Expected value" language shows up in theory and derivations (explaining why a model or estimator works), but the actual number crunched from your dataset (average income, average default rate) is the sample average formula in action — even though papers sometimes still describe it using "expected value" language when explaining the underlying model.

In one sentence: expected value is the language of theory and the true target you're estimating; sample average is what you actually calculate on real data, and the two converge to the same number as your data grows.

Wooldridge Appendix B.4

Features of Joint and Conditional Distributions · 13 August 2026 · intuition only, 11 pieces

Piece 1 — Covariance: do two things move together?

Imagine tracking two things every day for a month: ice cream sales and temperature. On hot days, ice cream sales tend to be high. On cold days, ice cream sales tend to be low. These two things move together — when one is above its own average, the other tends to be above its average too.

Covariance is just a number that captures this: do two variables tend to move in the same direction, or opposite directions, or no consistent pattern at all? A positive covariance means they move together (like ice cream and temperature). A negative covariance means they move oppositely (like umbrella sales and sunny days). A covariance near zero means no consistent pattern.

In one sentence: covariance is a single number that tells you whether two variables tend to move together, move oppositely, or show no real pattern at all.

Piece 2 — Independence guarantees zero covariance, but not the reverse

If two things are truly independent (like a coin flip and today's weather), their covariance will always be zero — unrelated things shouldn't show a consistent pattern.

But the reverse isn't true, and this is a real trap: two variables can have zero covariance and still be strongly related, just not in a straight line. Example: X = temperature, Y = how uncomfortable you feel. Both very cold and very hot days make you uncomfortable — a real, curved relationship. Because it curves instead of running in a straight line, the covariance can come out to zero even though X and Y are clearly connected.

In one sentence: independent variables always have zero covariance, but zero covariance doesn't prove independence — covariance only catches straight-line relationships and can miss curved ones entirely.

Piece 3 — Covariance's real weakness: the units problem

Covariance's size depends on what units you use, in a way that doesn't reflect how related two things really are. Example: measuring education and earnings, if earnings are in dollars you might get $\text{Cov} = 5,000$. Measure the exact same data in thousands of dollars instead, and covariance becomes $5,000 \times (1/1000) = 5$. Same real people, same real relationship — just a completely different-looking number, purely from the unit choice.

In one sentence: covariance's size can be misleading, because simply changing units changes the number, even though the real-world relationship hasn't changed at all.

Piece 4 — Correlation: fixing the units problem

Correlation takes covariance and divides it by the standard deviations of both variables — this division cancels out the units problem from Piece 3 completely. No matter what units you originally used, correlation always gives the same number.

Even better: correlation is always squeezed between -1 and $+1$. Close to $+1$ means a strong positive relationship, close to -1 means a strong negative relationship, close to 0 means little to no straight-line relationship. Example: the correlation between education and earnings might be $.15$ — and this stays exactly $.15$ whether earnings are measured in dollars or thousands of dollars.

In one sentence: correlation is covariance "cleaned up" — dividing by the standard deviations removes the units problem and always gives a number between -1 and $+1$ that's directly comparable across any study.

Piece 5 — The variance of a sum

Imagine a restaurant's Friday profit (X) and Saturday profit (Y). You might think total weekend variance is just Friday's variance

plus Saturday's — that's only true if Friday and Saturday are unrelated (uncorrelated).

Example: both nights average \$300 with a standard deviation of \$15 (variance = 225 each). If independent, total variance = $225+225 = 450$, giving a standard deviation of about \$21.21. But if the two nights were connected (a great Friday tends to mean a great weekend overall), the total variance would be bigger than just adding the two separately — the ups and downs reinforce each other instead of averaging out.

In one sentence: when two things are unrelated, their combined variance is just the sum of the two — but when they're connected, you also have to account for how they move together.

Piece 6 — This is exactly why the binomial's variance is so simple

A binomial variable is just many independent yes/no trials added together (Piece 1 idea from B.2) — like flipping a coin 10 times and counting heads. Each individual trial has its own small variance on its own. Because these trials are independent, Piece 5's rule applies directly: the total variance is just each trial's small variance, added up n times.

Worked example — the airline, 120 reservations, 85% show-up chance: one person's individual variance is $.85 \times .15 = .1275$. Since all 120 people are independent, just add this 120 times: $120 \times .1275 = 15.3$. Converting to standard deviation (more intuitive): the square root of 15.3 is about 3.9. This means the actual number of people showing up typically bounces around by about ± 3.9 people from the expected 102.

In one sentence: because a binomial is just many independent trials added together, and independent things' variances simply add, the binomial's total variance is just one trial's variance repeated and added n times.

Piece 7 — Conditional expectation: how does the average change depending on something else?

Instead of one single average for an entire population, conditional expectation asks a more specific question: what's the average outcome, GIVEN a specific condition? Example: instead of "what's the average wage of everyone," ask "what's the average wage of people with exactly 12 years of education," and separately, "with 16 years." These are genuinely different numbers — the average wage changes depending on how much education you're looking at.

In one sentence: conditional expectation asks "what's the average of Y , once you already know something specific about X " — instead of one number for everyone, you get a different average for each possible value of X .

Piece 8 — Turning this into a formula, and why it doesn't have to be a straight line

Rather than a separate average for every single education level, one simple formula can summarize the whole relationship. Example: $E(\text{WAGE} \mid \text{EDUC}) = 1.05 + 0.45 \times \text{EDUC}$ — start with a base wage of \$1.05, add 45 cents for every year of education. For 8 years: $1.05 + 0.45(8) = \$4.65/\text{hour}$. For 16 years: $1.05 + 0.45(16) = \$8.25/\text{hour}$. This is exactly what a regression equation represents — plug in any value of X , get the expected average of Y .

This relationship doesn't have to be a straight line, though. Example: price and quantity demanded, $E(\text{Quantity} \mid \text{Price}) = 10/\text{Price}$. At \$1, quantity = 10. At \$2, quantity = 5 (a drop of 5 for a \$1 increase). At \$5, quantity = 2. At \$10, quantity = 1 (a drop of only 1, for a \$5 increase). At low prices, a small price increase causes a huge drop in demand; at high prices, even a big increase barely moves demand — a genuinely curved relationship, unlike the wage example's fixed 45 cents per year.

In one sentence: a formula lets you plug in any X and instantly get the expected average of Y — and that relationship can be a straight line (fixed effect per unit, like wages) or a curve (a shrinking effect, like price and demand).

Piece 9 — A known fact behaves like a constant

Think about being told "how tall will your friend's future child be." With no information about the parents, this is genuinely uncertain. But if you're told "both parents are exactly 190cm tall," that height stops being a guess — it becomes a known fact you build your calculation on top of. The same logic applies to education and wages: once someone's education level is a known fact (say, 12 years), anything calculated purely from that number — squaring it, doubling it — is also no longer random, it's just

arithmetic on a known value.

This is exactly what "holding X fixed" means in every regression: once you're conditioning on a variable, you get to treat it as a plain, known constant, not something still uncertain that needs to be averaged over.

In one sentence: once something becomes a known fact instead of a guess — like the parents' height, or someone's education level — you build your calculation on top of it as a fixed number, exactly what "holding X constant" means in regression work.

Piece 10 — The law of iterated expectations: finding the overall average without averaging everyone

Here's the problem: you want one single number — the average wage for an entire city — but all you have is a rule (from Piece 8) that gives you each education group's average separately, not one number for the whole population at once. To combine every group's average into one overall figure the honest way, you'd need to know exactly how common each education level is, then carefully weight every group's average by that share of the population.

The law of iterated expectations offers a genuine shortcut around all of that: instead of weighting every group individually, you can simply take the population's *average* education level as one single number, and plug that one number straight into the same wage formula — as if it belonged to one typical person.

Concretely: using $E(\text{WAGE} \mid \text{EDUC}) = 4 + 0.60 \times \text{EDUC}$, and knowing the city's average education level is 11.5 years, you get $E(\text{WAGE}) = 4 + 0.60(11.5) = \$10.90/\text{hour}$ — the correct overall average wage for the entire city, found without ever looking at a single person's individual wage or building a table of every education level.

In one sentence: instead of averaging every education group separately and weighting each one, you can plug the population's average education directly into the wage formula, and it automatically gives you the correct overall average — a genuine shortcut, not an approximation.

Piece 11 — Conditional variance: a genuinely separate question from the average

Conditional expectation asks "what's typical, given a condition." Conditional variance asks a completely different question: "how much do things vary around that typical value, given a condition?" Picture two groups of families: low-income families mostly save a similar, small amount each month — not much room to vary. High-income families' savings are far more spread out — some save almost nothing, some save a lot, since they have more discretionary room to choose.

The formula $\text{Var}(\text{SAVINGS} \mid \text{INCOME}) = 400 + 0.25 \times \text{INCOME}$ is just this picture written as math: as income goes up, the number on the right gets bigger, meaning the spread genuinely grows with income — exactly matching the low-income/high-income picture.

Here's the important warning the textbook makes: knowing how the *average* changes tells you nothing about how the *spread* changes — you must check both separately. Suppose it turns out low-income and high-income families actually save about the same amount on average, say **\$200/month each**. Looking only at the average, you'd wrongly conclude "income doesn't matter for savings behavior." But checking the variance reveals something the average completely hid: high-income families' savings are far more spread out — some at \$0, some at \$1000 — even while matching the low-income group's average exactly.

In one sentence: the average and the variance can tell completely different stories about the same data, so checking only one of them — assuming "same average" means "same behavior" — can hide something important that only the other one would reveal.

Wooldridge Appendix B.5

The Normal and Related Distributions · 14 August 2026 · intuition first, 9 pieces

Piece 1 — The normal distribution: the bell curve

The normal distribution is the most common, most important shape in statistics — the classic bell curve. Most values cluster near the middle, with fewer and fewer values as you move further out to either side.

Real examples that genuinely follow this shape: human heights, human weights, test scores, county unemployment rates. A real example that does NOT: income, which is skewed — a small number of very high earners stretch the curve to one side. Taking the log of income often does produce something close to a normal bell curve; when this happens, we say the original variable has a lognormal distribution.

In one sentence: the normal distribution is the classic symmetric bell curve that many real variables (height, test scores) naturally follow, but skewed variables like income don't — though taking their log often does produce a normal shape.

Piece 2 — The standard normal distribution

Among all possible normal distributions, one special case has mean exactly 0 and variance exactly 1 — called the standard normal distribution, with the variable usually named Z. This is exactly the same Z from standardizing in B.3.

One important qualifier: standardizing always fixes the mean and variance, but only produces the actual standard normal bell SHAPE if the original variable was already normally shaped to begin with. Standardizing a skewed variable like income gives mean 0 and variance 1, but not the bell shape — just a shifted/rescaled version of whatever shape it already had.

In one sentence: the standard normal is the one specific normal bell curve with mean 0 and variance 1 — guaranteed only if the original variable was already normally shaped.

Piece 3 — Converting any normal variable into a standard normal, and why

There are thousands of different possible bell curves — one for adult heights, one for test scores, one for temperatures — each centered somewhere different, each "stretched" by a different amount. Calculating probabilities separately for every one of these would need endless separate tables.

The shortcut: shrink and shift any bell curve so it always looks exactly the same — centered at 0, same width every time. This is exactly what the Z-score formula does: $Z = (X - \mu) / \sigma$. Subtracting the mean shifts the curve to center 0; dividing by the standard deviation shrinks or stretches it to the same standard width. Once every problem is converted into this one shared shape, a single table can answer any of them.

Worked example: $X \sim \text{Normal}(3, 4)$ — mean 3, variance 4, so standard deviation 2. Want $P(X \leq 1)$.

$$Z = (1 - 3) / 2 = -1$$

Looking up $Z = -1$ in the standard table gives .159 — a 15.9% chance that X is 1 or less, found without ever needing a table specific to this particular bell curve.

Why this matters in real research: Z lets you turn a real, specific number into a judgment — "is this actually surprising, or just normal variation?" Example: if most U.S. counties have a loan default rate near 5%, but one county has 12%, converting that to a Z-score (say, 3 typical-steps above average) and looking it up might show only about 0.1% of counties would normally be this extreme by pure chance. Seeing it happen anyway is the signal something real is going on — not just random noise — the same logic as flipping 10 heads in a row: the rarity itself is what tells you to stop assuming "just luck" and start looking for a real cause.

In one sentence: converting any normal variable to the standard normal lets you use one shared probability table for every bell curve, and a low resulting probability is exactly the signal to stop assuming "chance" and start looking for a real explanation.

Piece 4 — A full worked example

$X \sim \text{Normal}(4, 9)$ — mean 4, variance 9, standard deviation 3. Question: $P(2 \leq X \leq 6)$?

$$Z \text{ for } 2: (2 - 4) / 3 = -0.67 \quad Z \text{ for } 6: (6 - 4) / 3 = 0.67$$

$$\text{Look up: } \Phi(0.67) = .749 \quad \Phi(-0.67) = .251$$

$$.749 - .251 = .498$$

Roughly a 49.8% chance — essentially a coin flip — that X falls between 2 and 6.

In one sentence: to find the probability of landing between two numbers, convert both to Z , look up both in the table, then subtract the smaller probability from the bigger one.

Piece 5 — Scaling and shifting a normal variable (Property Normal.2)

This connects directly to B.3: scaling ($\times a$) and shifting ($+b$) a variable changes its mean and variance in the same predictable ways as before (Properties E.2, VAR.2) — with one bonus: if X is normal, then $aX+b$ is ALSO normal. The bell shape survives.

Example: $X \sim \text{Normal}(1, 9)$. $Y = 2X + 3$.

$$\text{New mean: } 2(1) + 3 = 5 \quad \text{New variance: } 2^2 \times 9 = 36 \quad Y \sim \text{Normal}(5, 36)$$

In one sentence: scaling and shifting a normal variable follows the same B.3 rules for mean and variance, with the bonus that the result stays perfectly bell-shaped.

Piece 6 — The one case where zero correlation DOES mean independence (Property Normal.3)

Recall the B.4 trap: temperature and comfort had zero covariance but were still related, because the relationship was curved, not straight — zero covariance normally only rules out straight-line relationships, not curved ones.

Property Normal.3 removes this trap in one specific case: if **X and Y are BOTH shaped like normal bell curves, that curved-relationship trick from B.4 becomes mathematically impossible**. With genuinely normal variables, zero covariance means truly no relationship at all — straight or curved.

In one sentence: with normal variables, zero covariance always means genuinely unrelated — the sneaky curved-relationship trick from B.4 simply can't happen here.

Piece 7 — Combining independent normal variables, and why this matters for sample averages (Property Normal.4)

If you combine several independent normal variables — add them, multiply by numbers — the result is ALSO normal. Example: three independent variables X_1, X_2, X_3 , each $\sim \text{Normal}(\mu, \sigma^2)$. Define $W = X_1 + 2X_2 - 3X_3$.

$$\text{Mean: } \mu + 2\mu - 3\mu = 0$$

Variance uses the same B.4 "variance of a sum" rule, remembering Property VAR.2 (squaring each multiplier first):

$$\text{Var}(W) = \sigma^2 + (2)^2\sigma^2 + (-3)^2\sigma^2 = 14\sigma^2$$

So $W \sim \text{Normal}(0, 14\sigma^2)$ — still a genuine bell curve.

Why this matters for a sample average: **a plain average of many people (heights, wages, test scores) is just another specific way of "combining" normal variables** — add them all up, then multiply each by $1/n$. Since averaging is just a special case of combining normal variables, the result is guaranteed to still be normal too. This is exactly why nearly every statistical test later (confidence intervals, hypothesis tests) can rely on the sample average following a known, predictable normal shape — that guarantee is the foundation making those later tools work correctly.

In one sentence: combining independent normal variables always produces another normal variable, and a sample average is just one specific combination — which is exactly why sample averages are guaranteed to be normal too, the foundation for confidence intervals and hypothesis tests later.

Piece 8 — The Chi-Square distribution

Take several independent standard normal variables ($Z_1, Z_2, Z_3, \dots$), square each one, then add them all up. That sum follows a Chi-Square distribution. The number of terms added together is called the degrees of freedom (df).

Since everything is squared first, a Chi-Square variable can never be negative, and its shape isn't a symmetric bell curve — it's lopsided (skewed).

What the resulting number actually tells you: imagine predicting several things (say, sales in three different months), each prediction a little off from reality. Converting each error to a Z-score and just adding them could let positive and negative errors cancel out, hiding how far off you really were. Squaring first (like Chi-Square does) fixes this — every error becomes positive, so nothing can cancel. A small Chi-Square value means predictions were generally close to reality; a large value means they were, overall, quite far off.

In one sentence: Chi-Square is the sum of several squared, independent standard normal variables, and it tells you the total combined size of several errors at once — squaring first so positive and negative errors can't cancel each other out and hide the real problem.

Piece 9 — The t and F distributions (brief)

The t distribution is built by combining a standard normal (Z) with a Chi-Square variable. It looks similar to the normal bell curve, but more spread out, with fatter tails — extreme values happen a bit more often than the normal distribution predicts.

The F distribution is built by combining two Chi-Square variables, used later for testing whether multiple things are true at once in a regression.

Why these matter: the t distribution is the "workhorse of classical statistics and multiple regression analysis" — it shows up constantly when checking whether a single regression coefficient is statistically significant. F shows up when testing several coefficients together at once.

In one sentence: t and F are both built from combinations of normal and Chi-Square variables, and they're the specific tools used constantly later to test whether regression results are statistically meaningful.

Appendix C.1 - C.4 — What You Actually Need for Research

Practical version, not the derivations · 20 August 2026 · companion to the C.5 practical guide · take the mechanics as given facts, focus on application

C.1 — Populations, Parameters, and Random Sampling

1 Population, sample, parameter, estimate — the basic vocabulary

Population = the whole group you actually care about (the ocean). Sample = the smaller group you actually measured (the cup). Parameter = the true, real number that exists but can never be directly observed. Estimate = your best guess at it, based on your sample.

What to do: every time you read or write "the estimate of X is..." in a paper, you're reading a guess built from a sample, standing in for a true population value nobody can ever directly see.

2 Random sampling (i.i.d.)

A valid random sample means every observation is independent (one doesn't affect another) and identically distributed (all drawn from the same population) — like 5 random strangers' heights, not one basketball player mixed in with people off the street.

What to do: when judging whether a dataset is a fair sample, ask "are these observations genuinely independent, and all from the same population?" — if not, standard formulas and confidence intervals may not apply correctly.

C.2 — Judging an Estimator

3 Unbiased

An estimator is unbiased if, averaged across many repeated samples, it lands exactly on the true value — no systematic drift in either direction.

What to do: "unbiased" is a property of the METHOD over many hypothetical repeats, not a promise that your one actual result is close to the truth.

4 Sampling variance and efficiency

Two estimators can both be unbiased while one is far more reliable — tightly clustered guesses (low variance) beat wildly scattered ones (high variance), especially since real research only ever gets one actual sample, not many repeated tries. When comparing two unbiased estimators, the more efficient one is simply whichever has the smaller variance.

What to do: when a method claims to be unbiased, also ask how reliable it is on a single try — that's what separates a genuinely useful estimator from a technically-correct-but-unreliable one.

5 Mean Squared Error (MSE)

When comparing estimators where one isn't even unbiased, MSE combines both bias and variance into a single score, catching cases like "always guess zero" — which would look perfect on variance alone (never changes) but is actually a terrible estimator once bias is accounted for.

What to do: if a method is described as "biased but still useful," MSE is usually the reasoning behind that claim — a small bias can be an acceptable trade-off if it comes with much lower variance.

C.3 — What Happens With Larger Samples

6 Consistency and the Law of Large Numbers

Consistency means an estimator genuinely improves and gets closer to the truth as the sample grows — unlike unbiasedness (repeating the same-size sample many times), consistency is about growing one sample larger and larger. The Law of Large Numbers is the formal, proven statement of exactly this.

What to do: bigger datasets aren't just "nice to have" — they're mathematically guaranteed to bring your estimate closer to the truth, which is the actual justification for preferring large samples in empirical research.

7 The Central Limit Theorem (CLT)

Many different sample averages, plotted together, always settle into a bell-curve shape — even if the original underlying data (like incomes or default rates) isn't bell-shaped at all.

What to do: this single fact is why nearly every regression, t-test, and confidence interval you'll compute is allowed to use normal-distribution math, even on messy, non-normal real-world data — it's the quiet foundation under almost all standard statistical inference.

C.4 — Three Ways to Build an Estimator

8 Method of Moments, Maximum Likelihood, and Least Squares

Earlier appendices taught how to GRADE an estimator once someone hands you one (unbiased? consistent? efficient?). This section covers three different general "recipes" that actually PRODUCE a good estimator in the first place, rather than needing to guess-and-check from infinite possibilities.

Method of Moments: substitute the sample average wherever the true population mean would appear in a formula. Maximum Likelihood: pick whichever parameter value makes the data you actually observed most probable. Least Squares: pick whichever value makes the total squared prediction errors as small as possible.

These three different philosophies often arrive at the exact same estimator in practice — for example, the sample average ($\bar{Y}$) is simultaneously a Method of Moments estimator, a Maximum Likelihood estimator (for normal and Bernoulli data), and a Least Squares estimator, all at once.

What to do: when a paper mentions which method was used to estimate something (Method of Moments, MLE, OLS/Least Squares), recognize these as three different roads that frequently lead to the same destination — the choice matters less than knowing which one is being used and why.

Confidence Intervals — What You Actually Need for Research

Practical version, not the derivations · 20 August 2026 · take the mechanics as given facts, focus on application

1 What a confidence interval actually is

A point estimate (one number, like "training increases wages by 6.4%") never tells you how much to trust it. A confidence interval reports a RANGE instead ("the effect is likely between 4% and 9%"), which directly communicates how uncertain the guess is.

Take as given: the calculation itself (the formula, the t-distribution, the exact numbers) is settled statistical machinery — not something to verify or challenge at this stage.

What to do: whenever reporting an estimate in your own future work, report the interval alongside it, not just the single number.

2 The correct way to talk about "95% confidence"

Never say "there's a 95% chance the true value is in this specific interval" — once calculated, the interval is fixed, and the true value either is or isn't in it. The 95% describes the METHOD's success rate across many repeated samples, not this one specific result.

What to do: when writing or speaking about your own results, say "we are 95% confident" or "this is a 95% confidence interval," never "there's a 95% probability the true value falls here."

3 The single most important practical skill: does the interval exclude zero?

This is genuinely the most-used skill from this whole section. If your confidence interval for an effect does NOT contain zero, that's strong evidence the effect is real, not random noise. If zero IS inside the interval, you cannot confidently claim the effect exists at all — it could plausibly be nothing.

Real example: the job-training study's interval was $[-2.28, -0.02]$ — entirely negative, zero excluded — supporting a genuine effect. The hiring-discrimination study's interval was $[-.164, -.102]$ — again excluding zero, even at a stricter 99% level.

What to do: every time you compute a confidence interval in your own research, the first and most important check is simply "does this interval include zero?" — that single check is often the actual finding people care about.

4 Bigger sample = tighter, more trustworthy interval

The width of a confidence interval shrinks as sample size grows — the same "more data = more reliable" idea already covered for estimators generally. A study with 10,000 observations will report a much narrower, more precise interval than the same study with only 50.

What to do: when designing your own future research or judging someone else's, a very wide confidence interval is often a signal the sample size was too small to say much with confidence — worth noticing, not ignoring.

5 A confidence interval never fixes a bad research design

The job-training example showed a real, statistically solid interval (zero excluded) — but the study itself admitted it couldn't rule out other explanations for the improvement (other things could have changed at those firms that year, unrelated to the training).

What to do: a tight, zero-excluding confidence interval only tells you the NUMBER is reliable — it says nothing about whether your explanation for that number is actually correct. That's a separate question (identification strategy), not something a confidence interval can answer on its own.

Wooldridge Appendix C.6 & C.7

Hypothesis Testing & Notation · 21 August 2026 · the final section of Appendix C

1 What hypothesis testing is for

Some questions need a yes/no answer, not a range. Election example: Candidate A officially got 42%, but a sample of 100 voters shows 53% support him. Is this normal sampling noise, or evidence the count was wrong?

In one sentence: hypothesis testing exists for yes/no questions, using sample evidence to decide between two competing claims.

2 Null and alternative hypothesis

H0 (null) = the default, presumed-true claim (like a defendant's presumed innocence). H1 (alternative) = what you'd conclude if you reject H0. You need strong evidence to reject H0, not just a hunch.

In one sentence: the null hypothesis is presumed true until strong sample evidence says otherwise.

3 Type I and Type II errors, and significance level

Type I error: rejecting H0 when it was actually true (false accusation). Type II error: failing to reject H0 when it was actually false (missing real fraud). Significance level (α) = how much Type I error risk you accept, typically .05.

In one sentence: significance level is the accepted risk of a false positive, usually set at 5%.

4 The t-statistic — same tool as C.5, reused

Analogy: bolts normally weigh 50g with some variation. A sample averages 52.3g — is this normal, or alarming? Depends on how much bolts normally vary. The t-statistic answers exactly this: "how far is my result from expected, relative to normal variation?"

$$t = (\bar{y} - \mu_0) / \text{se}(\bar{y})$$

In one sentence: the t-statistic measures how far a result is from the null's claimed value, in standard-error units — the same raw gap can mean nothing or everything depending on normal variation.

5 Real research example — job training and defect rates

Holzer et al. (1993): 20 firms, change in scrap rate. H0: $\mu=0$ (no effect). Average change = -1.15, se = .54.

$$t = -1.15 / .54 = -2.13$$

Critical values: 5% level = -1.73, 1% level = -2.54. Since -2.13 is beyond -1.73, reject H0 at 5% level (genuine effect) — but not quite beyond -2.54, so fail to reject at the stricter 1% level.

In one sentence: the same result can be significant at one level (5%) and fall just short at a stricter one (1%).

6 p-values — the practical, everyday version

Real Stata example (regress price weight mpg):

weight		1.746559	.6413538	2.72	0.008
mpg		-49.51222	86.15604	-0.57	0.567

p-value = the probability of seeing a t-statistic this extreme, if the true effect were actually zero. weight: $p=.008$ (far below .05) -> reject H0, genuine effect. mpg: $p=.567$ (far above .05) -> fail to reject, no real evidence of an effect.

The practical rule to actually use: if p-value < .05, reject H0 (statistically significant). If p-value $\geq$.05, fail to reject (not significant). Focus on the p-value day-to-day — it's the direct, ready-to-use version of the t-statistic comparison, computed automatically by any statistics software.

In one sentence: $p < .05$ means reject the null; $p \geq .05$ means you can't.

7 Direction of the effect: read the coefficient, not the p-value

The rule: p-values are always positive and never show direction. Direction comes from the SIGN of the coefficient (or t-value) itself. Positive coefficient = positive relationship. Negative coefficient = negative relationship.

From the Stata output: weight = +1.75 (positive direction, as weight rises price rises). mpg = -49.5 (negative direction) — though not significant anyway ($p=.567$), so this direction isn't reliable evidence.

In one sentence: check the p-value for significance, check the coefficient's sign for direction — two separate pieces of information.

8 One-sided vs. two-sided tests

Two-sided = "is there any effect, in either direction" (the general, safer default — checks both directions equally, which is why Stata's column header is $P>|t|$, the absolute value of t). One-sided = "is there an effect specifically in ONE direction I've committed to in advance" (only appropriate when the opposite direction genuinely isn't a meaningful possibility worth testing, decided before seeing the data).

The rule: Stata defaults to two-sided ($P>|t|$). Two-sided is the safer, more honest default in most real research, since it can catch an unexpected effect in either direction. One-sided is reserved for narrow cases with a strong theoretical reason to only care about one direction, decided in advance — not chosen after seeing the data to make a marginal result look significant.

In one sentence: two-sided checks both directions equally (the safe default); one-sided commits to one direction in advance, and should only be used when the opposite direction genuinely isn't a meaningful possibility.

9 Confidence intervals and hypothesis tests are the same underlying check

The rule: if your hypothesized value (usually zero) falls OUTSIDE your confidence interval, that's the same as rejecting H_0 . If it falls INSIDE, that's the same as failing to reject.

Job-training example: 95% CI = [-2.28, -0.02]. Zero is outside this range -> reject $H_0: \mu=0$ at the 5% level, matching the t-statistic/p-value conclusion exactly.

In one sentence: often you don't need to calculate both separately — a confidence interval already answers the hypothesis test.

10 Practical significance vs. statistical significance — the biggest trap

Statistical significance (small p-value) only means an effect probably isn't zero. Practical significance means the effect is actually big enough to matter in real life. These are NOT the same.

Textbook example: widening a freeway, $n=900$ commuters, statistically significant ($p\approx.0005$) reduction in commute time — but only 3.6 minutes. Technically real, practically almost meaningless.

Fresh example: a weight-loss pill tested on 10,000 people shows a statistically significant ($p=.001$) difference of just 0.3kg over a year — real, but far too small to matter to anyone.

The rule: large samples can make even a practically trivial effect "statistically significant," because bigger samples shrink standard errors (C.2/C.3). Always check both: is it significant, AND is it actually big enough to matter?

In one sentence: a small p-value proves an effect probably isn't zero, but says nothing about whether that effect is large enough to actually matter.

11 Test consistency, and the $\hat{\theta}$ notation (C.7)

Test consistency: the same "gets closer to the truth with more data" idea from C.2/C.3, applied to hypothesis tests — exactly why very large samples can detect even practically trivial effects (Piece 10). Not a flaw, the test working as designed.

C.7 notation: from the main chapters onward, a "hat" symbol (θ , β , $\hat{\sigma}^2$) means BOTH the estimator and the estimate of that parameter — context tells you which, collapsing the estimator/estimate distinction into one simpler, universal symbol.

In one sentence: large samples detect even trivial effects because that's exactly what a consistent test is supposed

to do, and from now on a "hat" over any Greek letter just means "estimator or estimate of this parameter."

Questions asked during this session — short answers

Q: When would a researcher genuinely only care about one direction?

A: When the opposite direction isn't a realistic possibility worth testing — e.g. "does this drug reduce blood pressure," not "might it secretly raise it."

Q: Can "one direction" mean either positive or negative, or always positive?

A: Either — you choose the direction based on your actual question (one-sided-positive: "does X increase Y"; one-sided-negative: "does X decrease Y").

Q: Isn't one-sided testing risky, since it misses effects in the "wrong" direction (e.g. a drug secretly raising blood pressure)?

A: Yes, genuinely correct — this is a real, recognized limitation, which is exactly why two-sided tests are usually the safer default; one-sided tests should only be used with a strong, pre-existing theoretical reason, decided before seeing the data.